1. If $\begin{vmatrix} a+1 & 1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = 4$, then the value a is :

यदि $\begin{vmatrix} a+1 & 1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = 4$ तो a का मान है :

(A) 1 (B) – 1 (C) –2 (D) 0

Ans. D

Sol. $\begin{vmatrix} a+1 & 1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = 4$

$(a + 1)(1 + 1) – 1(1 – 1) + 1(1 + 1) = 4$

$2(a + 1) + 2 = 4$

$a + 1 = 1$

$a = 0$

2. If $\begin{vmatrix} x & y \\ 4 & 2 \end{vmatrix} = 7$ and $\begin{vmatrix} 2 & 3 \\ y & x \end{vmatrix} = 4$, then :

यदि $\begin{vmatrix} x & y \\ 4 & 2 \end{vmatrix} = 7$ तथा $\begin{vmatrix} 2 & 3 \\ y & x \end{vmatrix} = 4$, तो :

(A) $x = -3, y = -\frac{5}{2}$ (B) $x = -\frac{5}{2}, y = -3$ (C) $x = 3, y = \frac{5}{2}$ (D) $x = \frac{5}{2}, y = 3$

Ans. B

Sol. $2x – 4y = 7$

$2x – 3y = 4$

– + –

$-y = 3$

$y = -3$

$2x + 12 = 7$

$2x = -5$

$x = -\frac{5}{2}$

3. The value of $\begin{vmatrix} 5+i & -3i \\ 4i & 5-i \end{vmatrix}$ is :

$\begin{vmatrix} 5+i & -3i \\ 4i & 5-i \end{vmatrix}$ का मान है :

(A) 12 (B) 17 (C) 14 (D) 24

Ans. C

Sol. $(5 + i)(5 - i) + 12 (i)^2$

$25 + 1 - 12$

$26 - 12 = 14$

4. $\begin{vmatrix} \sec x & \sin x & \tan x \\ 0 & 1 & 0 \\ \tan x & \cot x & \sec x \end{vmatrix}$ is equal to :

$\begin{vmatrix} \sec x & \sin x & \tan x \\ 0 & 1 & 0 \\ \tan x & \cot x & \sec x \end{vmatrix}$ बराबर है :

(A) 0 (B) – 1 (C) 1 (D) None of these

Ans. C

Sol. $\sec^2 x - \tan^2 x = 1$

5. The value of $\frac{1}{x-y}\begin{vmatrix} 1 & 0 & 0 \\ 3 & x^3 & 1 \\ 5 & y^3 & 1 \end{vmatrix}$ is :

$\frac{1}{x-y}\begin{vmatrix} 1 & 0 & 0 \\ 3 & x^3 & 1 \\ 5 & y^3 & 1 \end{vmatrix}$ का मान है :

(A) $x+y$ (B) x^2-xy+y^2 (C) x^2+xy+y^2 (D) x^3-y^3

Ans. C

Sol. $\frac{1}{x-y}\left[1\left(x^3-y^3\right)\right]$

x^2+xy+y^2

6. The cofactors of $1, -2, -3$ and 4 in $\begin{vmatrix} 1 & -2 \\ -3 & 4 \end{vmatrix}$ are :

अवयव $1, -2, -3$ तथा 4 के सारणिक $\begin{vmatrix} 1 & -2 \\ -3 & 4 \end{vmatrix}$ के सहखण्ड होंगे :

(A) 4, 3, 2, 1 (B) $-4, 3, 2, -1$ (C) $4, -3, -2, 1$ (D) $-4, -3, -2, -1$

Ans. A

Sol. +4,3,2,1

7. The minors of the elements of the first row in the determinant $\begin{vmatrix} 2 & -1 & 4 \\ 4 & 2 & -3 \\ 1 & 1 & 2 \end{vmatrix}$ are :

सारणिक $\begin{vmatrix} 2 & -1 & 4 \\ 4 & 2 & -3 \\ 1 & 1 & 2 \end{vmatrix}$ की पहली पंक्ति के अवयवों के उपसारणिक हैं :

(A) 2, 7, 11 (B) 7, 11, 2 (C) 11, 2, 7 (D) 7, 2, 11

Ans. B

Sol. 7,11,2

8. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_2, B_2, C_2 are respectively cofactors of a_2, b_2, c_2 then $a_1A_2 + b_1B_2 + c_1C_2$ is equal to :

(A) $-\Delta$ (B) 0 (C) Δ (D) None of these

यदि $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ तथा A_2, B_2, C_2 क्रमशः a_2, b_2, c_2 के सहखण्ड हो, तो $a_1A_2 + b_1B_2 + c_1C_2$ बराबर है :

(A) $-\Delta$ (B) 0 (C) Δ (D) इनमें से कोई नहीं

Ans. B

Sol. On multiplying elements of any row with co-factor of corresponding elements of other row then added then the resultant sum will be zero.

$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$

9. If cofactor of 2x in the determinant $\begin{vmatrix} x & 1 & -2 \\ 1 & 2x & x-1 \\ x-1 & x & 0 \end{vmatrix}$ is zero, then x equals to :

यदि सारणिक $\begin{vmatrix} x & 1 & -2 \\ 1 & 2x & x-1 \\ x-1 & x & 0 \end{vmatrix}$ के अवयव $2x$ का सहखण्ड शून्य हो, तो x बराबर है :

(A) 0 (B) 2 (C) 1 (D) –1

Ans. C

Sol. $\begin{vmatrix} x & -2 \\ x-1 & 0 \end{vmatrix} \Rightarrow x \times 0 + 2(x-1) = 0$

$x - 1 = 0$

$x = 1$

10. The value of the determinant $\begin{vmatrix} a_1 & ma_1 & b_1 \\ a_2 & ma_2 & b_2 \\ a_3 & ma_3 & b_3 \end{vmatrix}$ is :

सारणिक $\begin{vmatrix} a_1 & ma_1 & b_1 \\ a_2 & ma_2 & b_2 \\ a_3 & ma_3 & b_3 \end{vmatrix}$ का मान है :

(A) 0 (B) $ma_1a_2a_3$ (C) $ma_1b_2a_2$ (D) $mb_1b_2b_3$

Ans. A

Sol. The elements of column C1 & C2 are proportional then the value of determinant is zero.

11. If $\Delta = \begin{vmatrix} a & 0 & 0 \\ b & c & a \\ c & a & b \end{vmatrix}$, then $\begin{vmatrix} p^2a & 0 & 0 \\ pb & c & a \\ pc & a & b \end{vmatrix}$ is equal to :

यदि $\Delta = \begin{vmatrix} a & 0 & 0 \\ b & c & a \\ c & a & b \end{vmatrix}$, तो $\begin{vmatrix} p^2a & 0 & 0 \\ pb & c & a \\ pc & a & b \end{vmatrix}$ बराबर है :

(A) $p\,\Delta$ (B) $p^2\,\Delta$ (C) $p^3\,\Delta$ (D) $2p\,\Delta$

Ans. B

Sol. $\begin{vmatrix} p^2a & 0 & 0 \\ pb & c & a \\ pc & a & b \end{vmatrix}$

$p\begin{vmatrix} pa & 0 & 0 \\ b & c & a \\ c & a & b \end{vmatrix} \Rightarrow p^2\Delta$

12. The value of the determinant $\begin{vmatrix} 1/a & 1 & bc \\ 1/b & 1 & ca \\ 1/c & 1 & ab \end{vmatrix}$ is equal to :

(A) abc (B) 1/abc (C) 0 (D) None of these

सारणिक $\begin{vmatrix} 1/a & 1 & bc \\ 1/b & 1 & ca \\ 1/c & 1 & ab \end{vmatrix}$ का मान है :

(A) abc (B) 1/abc (C) 0 (D) इनमें से कोई नहीं

Ans. C

Sol. $\begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$

$$\frac{1}{abc}\begin{vmatrix} 1 & a & abc \\ 1 & b & abc \\ 1 & c & abc \end{vmatrix} \Rightarrow \frac{abc}{abc}\begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0$$

13. If each row of a determinant of third order of value Δ is multiplied by 3, then the value of new determinant is :

यदि तीसरे क्रम के सारणिक जिसका मान Δ है, की प्रत्येक पंक्ति को 3 से गुणा किया जाये, तो प्राप्त नये सारणिक का मान है :

(A) Δ (B) 27Δ (C) 21Δ (D) 54Δ

Ans. B

Sol. If all the elements of a det. are multiplied by k then value of that det. gets multiplied by k^n where n is the order of det.

$= 3^3\Delta = 27\Delta$

14. The value of $\begin{vmatrix} a & ma+nx & x \\ b & mb+ny & y \\ c & mc+nz & z \end{vmatrix}$ is :

सारणिक $\begin{vmatrix} a & ma+nx & x \\ b & mb+ny & y \\ c & mc+nz & z \end{vmatrix}$ का मान है :

(A) $a+b+c$ (B) $x+y+z$

(C) $m(a+b+c)+n(x+y+z)$ (D) 0

Ans. D

Sol. $\begin{vmatrix} a & ma & x \\ b & mb & y \\ c & mc & z \end{vmatrix} + \begin{vmatrix} a & nx & x \\ b & ny & y \\ c & nz & z \end{vmatrix}$

$m(0)+n(0)=0$

15. The value of $\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$ is equal to :

$\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$ का मान है :

(A) a^3 (B) b^3 (C) c^3 (D) $a^3+b^3+c^3$

Ans. A

Sol. $\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$

$R_2 \to R_2 - 2R_1 \qquad R_3 \to R_3 - 3R_1$

$\begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{vmatrix}$

$a(a(7a+3b)-3a(2a+b))$

$a(7a^2+3ab-6a^2-3ab)$

$a+a^2=a^3$

16. The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is :

सारणिक $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ का मान है :

(A) $k(a+b)(b+c)(c+a)$ (B) $k\,abc\,(a^2+b^2+c^2)$

(C) $k(a-b)(b-c)(c-a)$ (D) $k(a+b-c)(b+c-a)(c+a-b)$

Ans. C

Sol. $k\begin{vmatrix} a & k^2+a^2 & 1 \\ b & k^2+b^2 & 1 \\ c & k^2+c^2 & 1 \end{vmatrix}$

$$k\left\{\begin{vmatrix} a & k^2 & 1 \\ b & k^2 & 1 \\ c & k^2 & 1 \end{vmatrix}+\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}\right\}$$

$k(a-b)(b-c)(c-a)$

17. If $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix}=0$, then value of x are :

यदि $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix}=0$ तो x का मान है :

(A) 0, a (B) 0, – a (C) a, – a (D) 0, 3a

Ans. D

Sol. $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$

$R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} 3a-x & 3a-x & 3a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$$

$$(3a-x)\begin{vmatrix} 1 & 1 & 1 \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$$

$C_2 \to C_2 - C_1 \qquad C_3 \to C_3 - C_1$

$$(3a-x)\begin{vmatrix} 1 & 0 & 0 \\ a-x & 2x & 0 \\ a-x & 0 & 2x \end{vmatrix} = 0$$

$(3a - x)\,(1(4x^2 - 0)) = 0$

$4x^2\,(3a - x) = 0$

$x = 0,\ x = 3a$

1. The value of the determinant $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is :

(A) abc (a – b) (b – c) (c – a) (B) (a – b) (b – c) (c – a) (a + b + c)
(C) (a – b) (b – c) (c – a) (ab + bc + ca) (D) None of these

सारणिक $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ का मान है :

(A) abc (a – b) (b – c) (c – a) (B) (a – b) (b – c) (c – a) (a + b + c)
(C) (a – b) (b – c) (c – a) (ab + bc + ca) (D) इनमें से कोई नहीं

Ans. C

Sol. $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$

$$\frac{1}{abc}\begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ abc & abc & abc \end{vmatrix}$$

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix} \Rightarrow (a-b)(b-c)(c-a)(ab+bc+ca)$$

2. If $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = k \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, then k is equal to :

यदि $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = k \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, तो k का मान है :

(A) 1 (B) 2 (C) 4 (D) 0

Ans. C

Sol. $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ a^2+1+2a & b^2+1+2b & c^2+1+2c \\ a^2+1-2a & b^2+1-2b & c^2+1-2c \end{vmatrix}$$

$R_2 \to R_2 + R_3$

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ 2(a^2+1) & 2(b^2+1) & 2(c^2+1) \\ a^2+1-2a & b^2+1-2b & c^2+1-2c \end{vmatrix}$$

$$2\begin{vmatrix} a^2 & b^2 & c^2 \\ a^2+1 & b^2+1 & c^2+1 \\ a^2+1-2a & b^2+1-2b & c^2+1-2c \end{vmatrix}$$

$R_3 \to R_3 - R_2$

$$2\begin{vmatrix} a^2 & b^2 & c^2 \\ a^2+1 & b^2+1 & c^2+1 \\ -2a & -2b & -2c \end{vmatrix}$$

$$-4\begin{vmatrix} a^2 & b^2 & c^2 \\ a^2+1 & b^2+1 & c^2+1 \\ a & b & c \end{vmatrix}$$

$$-4\left\{\begin{vmatrix} a^2 & b^2 & c^2 \\ a^2 & b^2 & c^2 \\ a & b & c \end{vmatrix}+\begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}\right\}$$

$= 4$

3. If x is real number such that $\begin{vmatrix} x+1 & x+2 & x+\alpha \\ x+2 & x+3 & x+\beta \\ x+3 & x+4 & x+\gamma \end{vmatrix} = 0$ then α, β, γ, are in :

(A) A.P. (B) G.P. (C) H.P. (D) None of these

यदि x वास्तविक हो ताकि $\begin{vmatrix} x+1 & x+2 & x+\alpha \\ x+2 & x+3 & x+\beta \\ x+3 & x+4 & x+\gamma \end{vmatrix} = 0$ तो α, β, γ होंगे :

(A) A.P. (B) G.P. (C) H.P. (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} x+1 & x+2 & x+\alpha \\ x+2 & x+3 & x+\beta \\ x+3 & x+4 & x+\gamma \end{vmatrix} = 0$

$R_2 \to R_2 - R_1 \qquad R_3 \to R_3 - R_1$

$$\begin{vmatrix} x+1 & x+2 & x+\alpha \\ 1 & 1 & \beta-\alpha \\ 2 & 2 & \gamma-\alpha \end{vmatrix} = 0$$

$C_2 - C_2 - C_1$

$$\begin{vmatrix} x+1 & 1 & x+\alpha \\ 1 & 0 & \beta-\alpha \\ 2 & 0 & \gamma-\alpha \end{vmatrix} = 0$$

$$-1(\gamma-\alpha-2\beta+2\alpha)=0$$

$$\alpha-2\beta+\gamma=0$$

$$\alpha+\gamma=2\beta \text{ A.P.}$$

4. The determinant $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2-bc & b^2-ca & c^2-ab \end{vmatrix}$ is equal to :

(A) 0 (B) 1 (C) – 1 (D) None of these

सारणिक $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2-bc & b^2-ca & c^2-ab \end{vmatrix}$ का मान है :

(A) 0 (B) 1 (C) – 1 (D) इनमें से कोई नहीं

Ans. A

Sol. $C_2 \to C_2 - C_1$ $C_3 \to C_3 - C_1$

$$\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2-bc & (b^2-ca)-(a^2-bc) & (c^2-ab)-(a^2-bc) \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2-bc & (a+b+c)(b-a) & (c-a)(a+b+c) \end{vmatrix}$$

$$(b-a)(c-a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 1 \\ a^2-bc & a+b+c & a+b+c \end{vmatrix}$$

$(b-a)(c-a)[(a+b+c)-(a+b+c)]=0$

5. The value of $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$ is :

(A) 0 (B) abc (C) $(a-b)(b-c)(c-a)$ (D) None of these

$\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$ का मान है :

(A) 0 (B) abc (C) $(a-b)(b-c)(c-a)$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$

$0(\ )-(a-b)(0-(c-a)(b-c))+(a-c)((b-a)(c-b)-0)$

$(a-b)(c-a)(b-c)+(a-c)(b-a)(c-b)$

$(a-b)(b-c)(c-a)-(a-b)(b-c)(c-a)=0$

6. If a, b, c are in A.P., then the value of $\begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix}$ equals :

यदि a, b, c समान्तर श्रेणी में हो, तो $\begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix}$ का मान बराबर है :

(A) 1 (B) 0 (C) 2a (D) 2x

Ans. B

Sol. $\begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix}$

$R_2 \to R_2 - R_1 \qquad R_3 \to R_3 - R_1$

$\begin{vmatrix} x+2 & x+3 & x+2a \\ 1 & 1 & 2b-2a \\ 2 & 2 & 2c-2a \end{vmatrix}$

$C_2 \to C_2 - C_1$

$\begin{vmatrix} x+2 & 1 & x+2a \\ 1 & 0 & 2b-2a \\ 2 & 0 & 2c-2a \end{vmatrix}$

$-1((2c-2a)-2(2b-2a))$

$-1(2(c-a)-4(b-a))$

$-1(2c-2a-4b+4a)$

$-1(2a-4b+2c)$

For A.P.

use $2b = a + c$

$-2a + 4b - 2c$

$4b - 2(a + c)$

$4b - 2(2b) = 0$

7. $\begin{vmatrix} 1+i & 1-i & i \\ 1-i & i & 1+i \\ i & 1+i & 1-i \end{vmatrix}$ (where $i = \sqrt{-1}$) equals :

$\begin{vmatrix} 1+i & 1-i & i \\ 1-i & i & 1+i \\ i & 1+i & 1-i \end{vmatrix}$ (जहाँ $i = \sqrt{-1}$) बराबर है :

(A) $7 + 4i$ (B) $7 - 4i$ (C) $4 + 7i$ (D) $4 - 7i$

Ans. C

Sol. $(1+i)(i(1-i)-(1+i)^2)-(1-i)((1-i)^2-i(1+i))+i((1+1)-i^2)$

$(1+i)(i+1-(1-1+2i))-(1-i)(1-1-2i-i-i^2)+i(2+1)$

$(1+i)(i+1-2i)-(1-1)(-3i+1)+3o$

$(1+i)(1-i)-(1-i)(1-3i)+3i$

$1+1-(1-3i-i-3)+3i$

$2+2+4i+3i \Rightarrow 4+7i$

8. The value of the determinant $\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix}$ is :

सारणिक $\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix}$ का मान है :

(A) 0 (B) $8x^2$ (C) 8 (D) –8

Ans. D

Sol. $\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix}$

$\begin{vmatrix} x^2+4-4x & x^2+1-2x & x^2 \\ x^2+1-2x & x^2 & x^2+1+2x \\ x^2 & x^2+1+2x & x^2+4+4x \end{vmatrix}$

$R_1 \to R_1 - R_2$

$\begin{vmatrix} 3-2x & 1-2x & -1-2x \\ x^2+1-2x & x^2 & x^2+1+2x \\ x^2 & x^2+2x+1 & x^2+4x+4 \end{vmatrix}$

$R_3 \to R_3 - R_2$

$$\begin{vmatrix} 3-2x & 1-2x & -1-2x \\ x^2+1-2x & x^2 & x^2+2x+1 \\ 2x-1 & 2x+1 & 2x+3 \end{vmatrix}$$

$R_1 \to R_1 + R_3$

$$\begin{vmatrix} 2 & 2 & 2 \\ x^2-2x+1 & x^2 & x^2+2x+1 \\ 2x-1 & 2x+1 & 2x+3 \end{vmatrix}$$

$$2\begin{vmatrix} 1 & 1 & 1 \\ x^2-2x+1 & x^2 & x^2+2x+1 \\ 2x-1 & 2x+1 & 2x+3 \end{vmatrix}$$

$C_2 \to C_2 - C_1$ $\quad$ $C_3 \to C_3 - C_1$

$$2\begin{vmatrix} 1 & 0 & 0 \\ x^2-2x+1 & 2x-1 & 4x \\ 2x-1 & 2 & 4 \end{vmatrix}$$

$$8\begin{vmatrix} 1 & 0 & 0 \\ x^2-2x+1 & 2x-1 & x \\ 2x-1 & 2 & 1 \end{vmatrix}$$

$8(2x - 1 - 2x)$

$= -8$

9. $\begin{vmatrix} 7579 & 7589 \\ 7581 & 7591 \end{vmatrix} =$

(A) 20 (B) – 2 (C) – 20 (D) 4

Ans. C

Sol. $\begin{vmatrix} 7579 & 7589 \\ 7581 & 7591 \end{vmatrix}$

$R_2 \to R_2 - R_1$

$\begin{vmatrix} 7579 & 7589 \\ 2 & 2 \end{vmatrix}$

$C_2 \to C_2 - C_1$

$\begin{vmatrix} 7579 & 10 \\ 2 & 0 \end{vmatrix} \Rightarrow -20$

10. If $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$ then x =

(A) 6 (B) 3 (C) 0 (D) None of these

यदि $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$ तो x =

(A) 6 (B) 3 (C) 0 (D) इनमें से कोई नहीं

Ans. C

Sol. $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$

$R_2 - R_2 - R_1$

$\begin{vmatrix} 3-x & -6 & 3 \\ -9+x & 9-x & 0 \\ 3 & 3 & -6-x \end{vmatrix} = 0$

$C_2 \to C_2 + C_1$

$$\begin{vmatrix} 3-x & -3-x & 3 \\ -9+x & 0 & 0 \\ 3 & 6 & -6-x \end{vmatrix} = 0$$

$(3-x)\{0-0\} + (x+3)\{(-6-x)(x-9)\} + 3(6(1-9+x)-0) = 0$

$-(x+3)\{(x+6)(x-9)\} + 18(x-9) = 0$

$-(x+3)\{x^2-9x+6x-54\} + 18x - 162 = 0$

$-(x+3)(x^2-3x-54) + 18x - 162 = 0$

$-\{x^3-3x^2-54x+3x^2-9x-162\} + 18x - 162 = 0$

$-x^3+3x^2+54x-3x^2+9x+162+18x-162 = 0$

$-x^3+81x = 0$

$-x(x^2-81) = 0$

$x = 0, x = 9, x = -9$

11. If $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = \lambda \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ then λ is equal to :

यदि $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = \lambda \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ तो λ का मान होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$

$R_2 \to R_2 - R_1$ $R_3 \to R_3 - R_1$

$\begin{vmatrix} a+b & b+c & c+a \\ c-a & a-b & b-c \\ c-b & a-c & b-a \end{vmatrix}$

$R_1 \to R_1 + R_2$

$$\begin{vmatrix} b+c & a+c & a+b \\ c-a & a-b & b-c \\ c-b & a-c & b-a \end{vmatrix}$$

$R_1 \to R_1 + R_3$

$$\begin{vmatrix} 2c & 2a & 2b \\ c-a & a-b & b-c \\ c-b & a-c & b-a \end{vmatrix}$$

$$2\begin{vmatrix} c & a & b \\ c-a & a-b & c-a \\ c-b & a-c & b-a \end{vmatrix}$$

$R_2 \to R_2 - R_1$

$$2\begin{vmatrix} c & a & b \\ -a & -b & -c \\ c-b & a-c & b-a \end{vmatrix}$$

$R_3 \to R_3 - R_1$

$$2\begin{vmatrix} c & a & b \\ -a & -b & -c \\ -b & -c & -a \end{vmatrix}$$

$$2\begin{vmatrix} c & a & b \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$2\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$\lambda = 2$

12. The determinant $\begin{vmatrix} ax & by & cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$ is equal to :

(A) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ x^2 & y^2 & z^2 \end{vmatrix}$ (B) $\begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix}$ (C) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ a^2 & b^2 & c^2 \end{vmatrix}$ (D) None of these

सारणिक $\begin{vmatrix} ax & by & cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$ का मान है :

(A) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ x^2 & y^2 & z^2 \end{vmatrix}$ (B) $\begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix}$ (C) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ a^2 & b^2 & c^2 \end{vmatrix}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\begin{vmatrix} ax & by & cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$

$$\frac{1}{yz \times zx \times xy}\begin{vmatrix} axyz & bzxy & cxy \\ x^2yz & y^2zx & z^2xy \\ yz & zx & xy \end{vmatrix}$$

$$\frac{x^2y^2z^2}{x^2y^2z^2}\begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix}$$

13. If the determinant $\begin{vmatrix} b-c & c-a & a-b \\ b'-c' & c'-a' & a'-b' \\ b''-c'' & c''-a'' & a''-b'' \end{vmatrix}$ is expressible as m $\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix}$, then the value of m is :

यदि सारणिक $\begin{vmatrix} b-c & c-a & a-b \\ b'-c' & c'-a' & a'-b' \\ b''-c'' & c''-a'' & a''-b'' \end{vmatrix}$ को m $\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix}$ रूप में व्यक्त किया जाये, तो m का मान होगा :

(A) – 1 (B) 0 (C) 1 (D) 2

Ans. B

Sol. $\begin{vmatrix} b-c & c-a & a-b \\ b'-c' & c'-a' & a'-b' \\ b''-c'' & c''-a'' & a''-b'' \end{vmatrix}$

$C_2 \to C_2 + C_1 \qquad C_3 \to C_3 + C_1$

$\begin{vmatrix} b-c & b-a & a-c \\ b'-c' & b'-a' & a'-c' \\ b''-c'' & b''-a'' & a''-c'' \end{vmatrix}$

$C_2 \to C_2 + C_3$

$\begin{vmatrix} b-c & b-c & a-c \\ b'-c' & b'-c' & a'-c' \\ b''-c'' & b''-c'' & a''-c'' \end{vmatrix}$

$C_1 \to C_1 - C_2$

$\begin{vmatrix} 0 & - & - \\ 0 & - & - \\ 0 & - & - \end{vmatrix}$

0

14. In a third order determinant each element of the first column consists of sum of two terms, each element of the second column consists of sum of three terms and each element of third column consists of sum of four terms, then it can be decomposed into n determinants, where n has the value :

यदि एक तीन कोटि के सारणिक के प्रथम स्तम्भ के सभी अवयव दो दो पदों के योग हैं, दूसरी स्तम्भ के सभी अवयव तीन तीन पदों के योग हैं तथा तीसरे स्तम्भ के सभी अवयव चार चार पदों के योग हैं तो इसे तीन कोटि के n सारणिकों के योग के रूप में लिखा जा सकता है जहाँ n बराबर है :

(A) 1 (B) 9 (C) 16 (D) 24

Ans. D

Sol. $\left|2 \text{ terms} \quad 3 \text{ terms} \quad 4 \text{ terms}\right|$

(d) 24

15. If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$ then $\sum_{r=1}^{n} S_r$ does not depends on :

(A) x (B) y (C) n (D) all of these

यदि $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$ तो $\sum_{r=1}^{n} S_r$ निम्न पर निर्भर नहीं है :

(A) x (B) y (C) n (D) उपरोक्त सभी

Ans. D

Sol. $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$

$$\sum_{r=1}^{n} S_r = \begin{vmatrix} 2(1+2+3+....n) & x & n(n+1) \\ 6(1^2+2^2+3^2+.....n^2)-n & y & n^2(2n+3) \\ 4(1^3+2^3+....n^3)-2n(1+2+3+....n) & z & n^3(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} \frac{2n(n+1)}{2} & x & n(n+1) \\ \frac{6n(n+1)(2n+1)}{6} & y & n^2(2n+3) \\ 4\left(\frac{n(n+1)}{2}\right)^2 - 2n\frac{n(n+1)}{2} & z & n^3(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} n(n+1) & x & n(n+1) \\ n(n+1)(2n+1)-n & y & n^2(2n+3) \\ n^3(n+1) & z & n^3(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} n(n+1) & x & n(n+1) \\ n^2(2n+3) & y & n^2(2n+3) \\ n^3(n+1) & z & n^3(n+1) \end{vmatrix} = 0$$

16. $\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$ is equal to :

$\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$ का मान है :

(A) $(x+p)(x+q)(x-p-q)$ (B) $(x-p)(x-q)(x+p+q)$

(C) $(x-p)(x-q)(x-p-q)$ (D) $(x+p)(x+q)(x+p+q)$

Ans. B

Sol. $\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$

$C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} x+p+q & p & q \\ x+p+q & x & q \\ x+p+q & q & x \end{vmatrix}$$

$$(x+p+q)\begin{vmatrix} 1 & p & q \\ 1 & x & q \\ 1 & q & x \end{vmatrix}$$

$R_2 \to R_2 - R_1$ $R_3 \to R_3 - R_1$

$$(x+p+q)\begin{vmatrix}1 & p & q\\ 0 & x-p & 0\\ 0 & q-p & x-q\end{vmatrix}$$

$(x + p + q)((x - p)(x - q) - 0)$

$(x + p + q)(x - p)(x - q)$

17. If [a] denotes the greatest integer less than or equal to a and $-1 \le x < 0$, $0 \le y < 1$, $1 \le z < 2$, then $\begin{vmatrix}[x]+1 & [y] & [z]\\ [x] & [y]+1 & [z]\\ [x] & [y] & [z]+1\end{vmatrix}$ is equal to :

(A) [x] (B) [y] (C) [z] (D) None of these

यदि [a] महत्तम पूर्णांक फलन $\le$ a तथा $-1 \le x < 0$, $0 \le y < 1$, $1 \le z < 2$ तो, $\begin{vmatrix}[x]+1 & [y] & [z]\\ [x] & [y]+1 & [z]\\ [x] & [y] & [z]+1\end{vmatrix}$ बराबर है :

(A) [x] (B) [y] (C) [z] (D) इनमें से कोई नहीं

Ans. C

Sol. $[z] = 1$

$[y] = 0$

$[x] = -1$

so,

$$= \begin{vmatrix}0 & 0 & 1\\ -1 & 1 & 1\\ -1 & 0 & 2\end{vmatrix}$$

$= 0(2 - 0) - 0(-2 + 1) + 1(0 + 1)$

$= 1$

18. If $\begin{vmatrix}x^2+x & x+1 & x-2\\ 2x^2+3x-1 & 3x & 3x-3\\ x^2+2x+3 & 2x-1 & 2x-1\end{vmatrix} = Ax - 12$, then the value of A is :

यदि $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax - 12$ तो A का मान है :

(A) 12 (B) 24 (C) –12 (D) –24

Ans. B

Sol. $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix}$

$R_2 \to R_2 - (R_1 + R_3)$

$\begin{vmatrix} x^2+x & x+1 & x-2 \\ -4 & 0 & 0 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax - 12$

$4((x+1)(2x-1) - (x-2)(2x-1)) = Ax - 12$

$4((2x^2 - x + 2x - 1) - (2x^2 - x - 4x + 2)) = Ax - 12$

$4(6x - 3) = Ax - 12$

$24x - 12 = Ax - 12$

$A = 24$

19. In a ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$ then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to :

एक ΔABC में, यदि $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$, तो $\sin^2 A + \sin^2 B + \sin^2 C$ का मान है :

(A) $\frac{9}{4}$ (B) $\frac{4}{9}$ (C) 1 (D) $3\sqrt{3}$

Ans. A

Sol. $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$

$R_2 \to R_2 - R_1$ $\qquad$ $R_3 \to R_3 - R_1$

$$\begin{vmatrix} 1 & a & b \\ 0 & c-a & a-b \\ 0 & b-a & c-b \end{vmatrix} = 0$$

$1\ \{(c-a)(c-b) - (a-b)(b-a)\} = 0$

$(c^2 - bc - ac + ab) + a^2 + b^2 - 2ab$

$a^2 + b^2 + c^2 - ab - bc - ac = 0$

$a^2 + b^2 + c^2 = ab + bc + ca$

$2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0$

$(a-b)^2 + (b-c)^2 + (c-a)^2 = 0$

$a = b \quad b = c \quad c = a$

$\theta = 60^\circ$ $\qquad$ $3\sin^2 A$

$$3 \times \frac{3}{4} = \frac{9}{4}$$

1. Let $px^4+qx^3+rx^2+sx+t=\begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$ be an identity, Where p, q, r, s, t are constants. Then the value of t equals :

माना $px^4+qx^3+rx^2+sx+t=\begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$ एक सर्वसमिका है। जहाँ p, q, r, s, t नियतांक है तो t का मान है:

(A) 0 (B) 1 (C) 2 (D) – 1

Ans. A

Sol. Hint : Put $x=0$

2. The solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix}=0$ is :

समीकरण $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix}=0$ के हलों का समुच्चय है :

(A) $\{1,2\}$ (B) $\{-1,-2\}$ (C) $\{1,-2\}$ (D) $\{-1,2\}$

Ans. D

Sol. $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix}=0$

$R_2 \to R_2-R_1$ $R_3 \to R_3-R_1$

$$\begin{vmatrix} 1 & 4 & 20 \\ 0 & -6 & -15 \\ 0 & 2x-4 & 5x^2-20 \end{vmatrix} = 0$$

$1(-6(5x^2-20)+15(2x-4))=0$

$-30x^2+120+30x-60=0$

$-30x^2+30x+60=0$

$-30(x^2-x-2)=0$

$x^2-x-2=0$

$x^2-2x+1x-2-=0$

$x(x-2)+1(x-2)=0$

$(x-2)(x+1)=0$

$x=-1, 2$

3. Given $x=-9$ is a root of the equation, $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$, the other roots are :

दिया गया है कि $x=-9$ समीकरण $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ का एक मूल है तो अन्य मूल ज्ञात कीजिये :

(A) $2, -7$ (B) $-7, 2$ (C) $-2, -7$ (D) $2, 7$

Ans. D

Sol. $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix}$

$x(x^2-12)-3(2x-14)+7(12-7x)=0$

$x^3-12x-6x+42+84-49x=0$

$x^3-67x+126=0$

$(x+9)(x^2-9x+14)=0$

$x^2-7x-2x+14=0$

$x(x-7)-2(x-7)=0$

$(x-7)(x-2)=0$

$x=2,7$

4. If $\begin{vmatrix}1 & a & bc\\ 1 & b & ca\\ 1 & c & ab\end{vmatrix}=\lambda\begin{vmatrix}a^2 & b^2 & c^2\\ a & b & c\\ 1 & 1 & 1\end{vmatrix}$, then $\lambda=$

यदि $\begin{vmatrix}1 & a & bc\\ 1 & b & ca\\ 1 & c & ab\end{vmatrix}=\lambda\begin{vmatrix}a^2 & b^2 & c^2\\ a & b & c\\ 1 & 1 & 1\end{vmatrix}$, तो $\lambda=$

(A) 1 (B) – 1 (C) 2 (D) – 3

Ans. B

Sol. $\begin{vmatrix}1 & a & bc\\ 1 & b & ca\\ 1 & c & ab\end{vmatrix}$

$\frac{1}{abc}\begin{vmatrix}a & a^2 & abc\\ b & b^2 & abc\\ c & c^2 & abc\end{vmatrix}$

$\begin{vmatrix}a & a^2 & 1\\ b & b^2 & 1\\ c & c^2 & 1\end{vmatrix}$

$\begin{vmatrix}a & b & c\\ a^2 & b^2 & c^2\\ 1 & 1 & 1\end{vmatrix}$

$$-\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

$\lambda = -1$

5. $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix} =$

(A) 0 (B) $\sum a^2(b-c)$ (C) $2\sum a^2(b-c)$ (D) $-2\sum ab(a-b)$

Ans. A

Sol. $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$

$$\begin{vmatrix} a & a^2 & a^3 - abc \\ b & b^2 & b^3 - abc \\ c & c^2 & c^3 - abc \end{vmatrix}$$

$$\begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} - abc \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

$$abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} - abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$= 0$

6. If x, y, z are positive, then $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix} =$

(A) 0 (B) 1 (C) – 1 (D) None

यदि x, y, z धनात्मक है, तो $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix} =$

(A) 0 (B) 1 (C) – 1 (D) कोई नहीं

Ans. A

Sol. Do $R_1 \to \log x R_1, R_2 \to \log y R_2, R_3 \to \log z R_3$

we get $\frac{1}{\log x \log y \log z}\begin{vmatrix} \log x & \log y & \log z \\ \log x & \log y & \log z \\ \log x & \log y & \log z \end{vmatrix} = 0$

7. The number of real values of x satisfying $\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 7x-2 & 17x+6 & 12x-1 \end{vmatrix} = 0$ is :

(A) 3 (B) 0 (C) more than 3 (D) 1

$\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 7x-2 & 17x+6 & 12x-1 \end{vmatrix} = 0$ को संतुष्ट करने वाले x के वास्तविक मानों की संख्या होगी :

(A) 3 (B) 0 (C) 3 से अधिक (D) 1

Ans. C

Sol. Use $R_3 \to R_3 - (2R_2 + 3R_1)$ we get $\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 0 & 0 & 0 \end{vmatrix}$ which is equal to zero for all x.

8. The sum of infinite series $\begin{vmatrix} 1 & 2 \\ 6 & 4 \end{vmatrix} + \begin{vmatrix} \frac{1}{2} & 2 \\ 2 & 4 \end{vmatrix} + \begin{vmatrix} \frac{1}{4} & 2 \\ \frac{2}{3} & 4 \end{vmatrix} + \ldots\ldots$ is :

अनन्त श्रेढी $\begin{vmatrix} 1 & 2 \\ 6 & 4 \end{vmatrix} + \begin{vmatrix} \frac{1}{2} & 2 \\ 2 & 4 \end{vmatrix} + \begin{vmatrix} \frac{1}{4} & 2 \\ \frac{2}{3} & 4 \end{vmatrix} + \ldots\ldots$ का योग है :

(A) –10 (B) 0 (C) 10 (D) ∞

Ans. A

Sol. $\begin{vmatrix} 1+\frac{1}{2}+\frac{1}{4}+\ldots. & 2 \\ 6+2+\frac{2}{3}+\ldots. & 4 \end{vmatrix}$

$\begin{vmatrix} \frac{1}{1-\frac{1}{2}} & 2 \\ \frac{6}{1-\frac{1}{3}} & 4 \end{vmatrix}$

$\begin{vmatrix} \frac{2}{1} & 2 \\ 9 & 4 \end{vmatrix}$

$\Rightarrow 8 - 18 = -10$

9. $\begin{vmatrix} 2^3 & 3^3 & 3.2^2+3.2+1 \\ 3^3 & 4^3 & 3.3^2+3.3+1 \\ 4^3 & 5^3 & 3.4^2+3.4+1 \end{vmatrix}$ is equal to :

(A) 0 (B) 1 (C) 92 (D) None of these

$\begin{vmatrix} 2^3 & 3^3 & 3.2^2+3.2+1 \\ 3^3 & 4^3 & 3.3^2+3.3+1 \\ 4^3 & 5^3 & 3.4^2+3.4+1 \end{vmatrix}$ बराबर है :

(A) 0 (B) 1 (C) 92 (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} 2^3 & 3^3 & 19 \\ 3^3 & 4^3 & 37 \\ 4^3 & 5^3 & 61 \end{vmatrix}$

$\begin{vmatrix} 2^3 & 3^3 & 3^3-2^3 \\ 3^3 & 4^3 & 4^3-3^3 \\ 4^3 & 5^3 & 5^3-4^3 \end{vmatrix}$

$\begin{vmatrix} 2^3 & 3^3 & 3^3 \\ 3^3 & 4^3 & 4^3 \\ 4^3 & 5^3 & 5^3 \end{vmatrix} - \begin{vmatrix} 2^3 & 3^3 & 2^3 \\ 3^3 & 4^3 & 3^3 \\ 4^3 & 5^3 & 4^3 \end{vmatrix}$

$0 - 0 = 0$

10. If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = k\,(a+b+c)^3$, then k =

यदि $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = k\,(a+b+c)^3$, तो k =

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

$R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$(a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$C_2 \to C_2 - C_1 \qquad C_3 \to C_3 - C_1$

$$(a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix}$$

$$(a+b+c)^3\begin{vmatrix} 1 & 0 & 0 \\ 2b & -1 & 0 \\ 2c & 0 & -1 \end{vmatrix}$$

$(a + b + c)^3 (1)$

$k = 1$

11. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$ then $f(100) =$

यदि $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$ तो $f(100) =$

(A) 0 (B) 100 (C) 1 (D) –100

Ans. A

Sol. $f(x)=x(x+1)\begin{vmatrix} 1 & 1 & 1 \\ 2x & x-1 & x \\ 3x(x-1) & (x-1)(x-2) & x(x-1) \end{vmatrix}$

$C_2 \to C_2 - C_1 \qquad C_3 \to C_3 - C_1$

$$f(x)=x(x+1)\begin{vmatrix} 1 & 0 & 0 \\ 2x & -(x+1) & -x \\ 3x(x-1) & -2(x^2-1) & -2(x)(x-1) \end{vmatrix}$$

$$=x(x+1)(x+1)x\begin{vmatrix} 1 & 0 & 0 \\ 2x & -1 & -1 \\ 3x(x-1) & -2(x-1) & -2(x)(x-1) \end{vmatrix}$$

$$f(x)=x^2(x+1)^2(x-1)\begin{vmatrix} 1 & 0 & 0 \\ 2x & -1 & -1 \\ 3x(x-1) & -2 & -2 \end{vmatrix}$$

$f(x)=0$

$f(100)=0$

12. $\begin{vmatrix} 2xy & x^2 & y^2 \\ x^2 & y^2 & 2xy \\ y^2 & 2xy & x^2 \end{vmatrix} =$

(A) $(x^3+y^3)^2$ (B) $(x^2+y^2)^3$ (C) $-(x^2+y^2)^3$ (D) $-(x^3+y^3)^2$

Ans. D

Sol. $2xy(x^2y^2-4x^2y^2)-x^2(x^4-2xy^3)+y^2(2x^3y-y^4)$

$2x^3y^3-8x^3y^3-x^6+2x^3y^3+2x^3y^3-y^6$

$-x^6-y^6-2x^3y^3$

$-(x^6+y^6+2x^3y^3)$

$-((x^3)^2+(y^3)^2+2x^3y^3)$

$-(x^3+y^3)^2$

1. If $A = (a_{ij})$ is a 4×4 matrix and c_{ij} is the co-factor of the element a_{ij} in Det (A), then the expression

$a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} + a_{14}c_{14}$ equals :

यदि $A = (a_{ij})$ एक 4×4 क्रम का मेट्रिक्स है तथा c_{ij} सारणिक (A) में अवयव a_{ij} का सहखण्ड है, तो व्यंजक $a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} + a_{14}c_{14}$ बराबर है :

(A) 0 (B) -1 (C) 1 (D) Det. (A)

Ans. D

Sol. By multiplying elements of a row with corresponding cofactors and then added, resultant sum will be equal to the value of det.

2. If $D_r = \begin{vmatrix} r & x & n(n+1)/2 \\ 2r-1 & y & n^2 \\ 3r-2 & z & n(3n-1)/2 \end{vmatrix}$, then $\sum_{r=1}^{n} D_r$ is equal to :

(A) $\frac{1}{6}n(n+1)(2n+1)$ (B) $\frac{1}{4}n^2(n+1)^2$ (C) 0 (D) None of these

यदि $D_r = \begin{vmatrix} r & x & n(n+1)/2 \\ 2r-1 & y & n^2 \\ 3r-2 & z & n(3n-1)/2 \end{vmatrix}$, तो $\sum_{r=1}^{n} D_r$ का मान होगा :

(A) $\frac{1}{6}n(n+1)(2n+1)$ (B) $\frac{1}{4}n^2(n+1)^2$ (C) 0 (D) इनमें से कोई नहीं

Ans. C

Sol. $Dr = \begin{bmatrix} r & x & \frac{n(n+1)}{2} \\ 2r-1 & y & n^2 \\ 3r-2 & z & \frac{n(3n-1)}{2} \end{bmatrix}$

$$\sum_{r=1}^{n} Dr = \begin{vmatrix} \frac{n(n+1)}{2} & x & \frac{n(n+1)}{2} \\ 2\left(\frac{n(n+1)}{2}\right)-n & y & n^2 \\ 3\left(\frac{n(n+1)}{2}\right)-2n & z & n\frac{(3n-1)}{2} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{n(n+1)}{2} & x & \frac{n(n+1)}{2} \\ n^2 & y & n^2 \\ \frac{n(3n-1)}{2} & z & \frac{n(3n-1)}{2} \end{vmatrix} = 0$$

3. The value of $\begin{vmatrix} a & b+c & a^3 \\ b & c+a & b^3 \\ c & a+b & c^3 \end{vmatrix}$ is :

(A) $(a-b)(b-c)(c-a)$ (B) $abc(a-b)(b-c)(c-a)$

(C) $-(a+b+c)^2(a-b)(b-c)(c-a)$ (D) None of these

$\begin{vmatrix} a & b+c & a^3 \\ b & c+a & b^3 \\ c & a+b & c^3 \end{vmatrix}$ का मान है :

(A) $(a-b)(b-c)(c-a)$ (B) $abc(a-b)(b-c)(c-a)$

(C) $-(a+b+c)^2(a-b)(b-c)(c-a)$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\begin{vmatrix} a & b+c & a^3 \\ b & c+a & b^3 \\ c & a+b & c^3 \end{vmatrix}$

$C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} a+b+c & b+c & a^3 \\ a+b+c & c+a & b^3 \\ a+b+c & a+b & c^3 \end{vmatrix}$$

$$(a+b+c)\begin{vmatrix} 1 & b+c & a^3 \\ 1 & c+a & b^3 \\ 1 & a+b & c^3 \end{vmatrix}$$

$R_2 \to R_2 - R_1$ $\qquad$ $R_3 \to R_3 - R_1$

$$(a+b+c)\begin{vmatrix} 1 & b+c & a^3 \\ 0 & a-b & b^3-a^3 \\ 0 & a-c & c^3-a^3 \end{vmatrix}$$

$$(a+b+c)(b-a)(c-a)\begin{vmatrix} 1 & b+c & a^3 \\ 0 & -1 & a^2+b^2+ab \\ 0 & -1 & c^2+a^2+ca \end{vmatrix}$$

$(a + b + c)\,(b - a)\,(c - a)\,[-c^2 - a^2 - ac + a^2 + b^2 + ab]$

$(a + b + c)\,(b - a)\,(c - a)\,[(b + c)\,(b - c) + a\,(b - c)]$

$-(a + b + c)^2\,(a - b)\,(b - c)\,(c - a)$

4. If $A + B + C = \pi$, then $\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ equals :

(A) 0 (B) $2\sin B \tan A \cos C$ (C) 1 (D) None of these

यदि $A + B + C = \pi$, then $\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ बराबर है :

(A) 0 (B) $2\sin B \tan A \cos C$ (C) 1 (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} 0 & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ -\cos C & -\tan A & 0 \end{vmatrix}$

$0(\ ) - \sin B (0 + \tan \cos C) + \cos C (\tan A \sin B)$

$-\sin B \tan A \cos C + \cos C \tan A \sin B$

5. Given the system of equations $px + y + z = 1, x + py + z = p, x + y + pz = p^2$, then for what value of p does this system have no solution :

समीकरण निकाय $px + y + z = 1, x + py + z = p, x + y + pz = p^2$ के लिऐ p का मान क्या हो, ताकि निकाय का कोई हल विद्यमान न हो :

(A) 0 (B) –2 (C) 1 (D) 0, 1

Ans. B

Sol. $px + y + z = 1$

$x + py + z = p$

$x + y + pz = p^2$

$$D \equiv \begin{vmatrix} p & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & p \end{vmatrix} = 0$$

$D \equiv p(p^2 - 1) - 1(p - 1) + 1\ (1 - p) = 0$

$p^3 - p - p + 1 + 1 - p = 0$

$p^3 - 3p + 2 = 0$

$p^2 (p - 1) + p (p - 1) - 2 (p - 1)$

$(p - 1) (p^2 + p - 2) = 0$

$(p - 1)^2 (p + 2) = 0$

$p = 1, -2$

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ p & p & 1 \\ p^2 & 1 & p \end{vmatrix}$$

for $p = 1$, $D_1 = 0$

$p = -2$, $D_1 \neq 0$

$$D_2 = \begin{vmatrix} p & 1 & 1 \\ 1 & p & 1 \\ 1 & p^2 & p \end{vmatrix}$$

$$D_3 = \begin{vmatrix} p & 1 & 1 \\ 1 & p & p \\ 1 & 1 & p^2 \end{vmatrix}$$

$p = -2$

6. The value of k for which the set of equations $3x + ky - 2z = 0$, $x + ky + 3z = 0$ and $2x + 3y - 4z = 0$ has a non – trivial solution is :

k का मान ताकि समीकरण निकाय $3x + ky - 2z = 0$, $x + ky + 3z = 0$ एवं $2x + 3y - 4z = 0$ का एक अनिरर्थक हल हो :

(A) 15 (B) 16 (C) 31/2 (D) 33/2

Ans. D

Sol. $3x + 5y - 2z = 0$

$x + ky + 3z = 0$

$2x + 3y - 4z = 0$

$$D = \begin{vmatrix} 3 & k & -2 \\ 1 & k & 3 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$3(-4k - 9) - k(-4 - 6) - 2(3 - 2k) = 0$

$-12k - 27 + 10k - 6 + 4k = 0$

$2k - 33 = 0$

$$k = \frac{33}{2}$$

7. If $\Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$ then Δ_1 is equal to :

(A) $2\Delta_2$ (B) Δ_2 (C) $-\Delta_2$ (D) None of these

यदि $\Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$ तथा $\Delta_2 = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$ तो Δ_1 बराबर है :

(A) $2\Delta_2$ (B) Δ_2 (C) $-\Delta_2$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$

$\Delta_1 = \begin{vmatrix} a & x & p \\ b & y & q \\ x & z & r \end{vmatrix}$

$\Delta_1 = -\begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix}$

$\Delta_1 = -\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

$\Delta_1 = \Delta_2$

8. If $D_p = \begin{vmatrix} p & 15 & 8 \\ p^2 & 35 & 9 \\ p^3 & 25 & 10 \end{vmatrix}$, then $D_1 + D_2 + D_3 + D_4 + D_5$ is equal to :

(A) 0 (B) 25 (C) 625 (D) None of these

यदि $D_p = \begin{vmatrix} p & 15 & 8 \\ p^2 & 35 & 9 \\ p^3 & 25 & 10 \end{vmatrix}$, तो $D_1 + D_2 + D_3 + D_4 + D_5$ बराबर है :

(A) 0 (B) 25 (C) 625 (D) इनमें से कोई नहीं

Ans. D

Sol. $D_p = \begin{vmatrix} p & 15 & 8 \\ p^2 & 35 & 9 \\ p^3 & 25 & 10 \end{vmatrix}$

$$D_p = 5p \begin{vmatrix} 1 & 3 & 8 \\ p & 7 & 9 \\ p^2 & 5 & 10 \end{vmatrix}$$

$$D_p = 5p\left(1(70-45) - 3\left(10p - 9p^2\right) + 8\left(5p - 7p^2\right)\right)$$

$D_p = 5p\,(25 - 30p + 27p^2 + 40p - 56p^2)$

$D_p = 5p(25 + 10p - 29p^2)$

$D_p = 5p(-29p^2 + 10p + 25)$

$D_1 = 5(-29 + 10 + 25) = 30$

$D_2 = 10(-116 + 20 + 25) = 10 \times (-71) = -710$

$D_3 = 15(-261 + 30 + 25) = 15 \times (-206) = -3090$

$D_4 = 20(-454 + 40 + 25) = 15 \times (-399) = -7980$

$D_5 = 25\,(-725 + 50 + 25) = 25 \times (-650) = -16250$

$D_1 + D_2 + D_3 + D_4 + D_5 = -28000$

9. If $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (x-y)(y-z)(z-x)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$, then n is equals to :

यदि $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (x-y)(y-z)(z-x)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ तो n बराबर है :

(A) 1 (B) –1 (C) 2 (D) –2

Ans. B

Sol. $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = x^n y^n z^n \begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix}$

$= (x-y)(y-z)(z-x)(xy+zy+zx)\,x^n y^n z^n$ at $n=-1$ पर

$= (x-y)(y-z)(z-x)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$

10. If α, β and γ are the roots of $x^3+ax^2+b=0$, then the value of $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is equals to :

यदि α, β तथा γ समीकरण $x^3+ax^2+b=0$, के मूल हो, तो $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ का मान है :

(A) $-a^3$ (B) a^3-3b (C) a^3 (D) a^2-3b

Ans. C

Sol. $x^3+ax^2+b=0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$

$\alpha+\beta+\gamma=-a$

$\alpha\beta+\beta\gamma+\gamma\alpha=0$

$\alpha\beta\gamma=-b$

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$$

$$R_1 \to R_1 + R_2 + R_3$$

$$\begin{vmatrix} \alpha+\beta+\gamma & \alpha+\beta+\gamma & \alpha+\beta+\gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$$

$$(\alpha+\beta+\gamma)\begin{vmatrix} 1 & 1 & 1 \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$$

$C_2 \to C_2 - C_1$ $\qquad$ $C_3 \to C_3 - C_1$

$$(\alpha+\beta+\gamma)\begin{vmatrix} 1 & 0 & 0 \\ \beta & \gamma-\beta & \alpha-\beta \\ \gamma & \alpha-\gamma & \beta-\gamma \end{vmatrix}$$

$$(\alpha+\beta+\gamma)(\gamma-\beta)(\beta-\gamma)-(\alpha-\gamma)(\alpha-\beta)$$

$$(-a)\left(\beta\gamma-\gamma^2-\beta^2+\beta\gamma-\alpha^2+\alpha\beta+\alpha\gamma-\beta\gamma\right)$$

$$(-a)\left(-\left(\alpha^2+\beta^2+\gamma^2-\alpha\beta-\beta\gamma-\gamma\alpha\right)\right)$$

$$a\left(\alpha^2+\beta^2+\gamma^2-2\alpha\beta-2\beta\gamma-2\gamma\alpha\right)+\alpha\beta+\beta\gamma+\gamma\alpha$$

$$a\left((\alpha+\beta+\gamma)^2-0\right)$$

$= a^3$

11. If A, B and C are the angles of a triangle and $\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A+\sin^2 A & \sin B+\sin^2 B & \sin C+\sin^2 C \end{vmatrix} = 0$, then the

triangle ABC is :

(A) isosceles (B) equilateral

(C) right angled isosceles (D) None of these

यदि A, B तथा C एक त्रिभुज के कोण है तथा $\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A+\sin^2 A & \sin B+\sin^2 B & \sin C+\sin^2 C \end{vmatrix} = 0$, तो त्रिभुज ABC है :

(A) समद्विबाहु त्रिभुज (B) समबाहु त्रिभुज

(C) समकोणीय समद्विबाहु त्रिभुज (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A+\sin^2 A & \sin B+\sin^2 B & \sin C+\sin^2 C \end{vmatrix} = 0$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A & \sin B & \sin C \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin^2 A & \sin^2 B & \sin^2 C \end{vmatrix} = 0$$

$R_2 \to R_2 - R_3$ $R_2 \to R_2 - R_1$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ \sin A & \sin B & \sin C \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ \sin A & \sin B & \sin C \\ \sin^2 A & \sin^2 B & \sin^2 C \end{vmatrix}$$

$0 + (\sin A - \sin B)(\sin B - \sin C)(\sin C - \sin A) = 0$

$\sin A = \sin B$ or $\sin B = \sin C$ or $\sin C = \sin A$

$A = B$ or $B = C$ or $C = A$

12. Let $D_r = \begin{vmatrix} a & 2^r & 2^{16}-1 \\ b & 3(4^r) & 2(4^{16}-1) \\ c & 7(8^r) & 4(8^{16}-1) \end{vmatrix}$, then the value of $\sum_{k=1}^{16} D_r$ is equals to :

(A) 0 (B) a + b + c (C) ab + bc + ca (D) None of these

माना कि $D_r = \begin{vmatrix} a & 2^r & 2^{16}-1 \\ b & 3\left(4^r\right) & 2\left(4^{16}-1\right) \\ c & 7\left(8^r\right) & 4\left(8^{16}-1\right) \end{vmatrix}$ है, तो $\sum_{k=1}^{16} D_r$ का मान है :

(A) 0 (B) a + b + c (C) ab + bc + ca (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} a & 2^r & 2^{16}-1 \\ b & 3\left(4^r\right) & 2\left(4^{16}-1\right) \\ c & 7\left(8^r\right) & 4\left(8^{16}-1\right) \end{vmatrix}$

$2^r = 2 + 2^2 + 2^3 + \ldots\ldots 2^{16} = \dfrac{2\left(2^{16}-1\right)}{2-1} = 2\,(2^{16}-1)$

$3 \cdot 4^r = 3 \cdot (4^1 + 4^2 + \ldots\ldots 4^{16}) = 3 \cdot \dfrac{4\left(4^{16}-1\right)}{4-1}$

$7 \cdot 8^r = 7(8^1 + 8^2 + \ldots\ldots + 8^{16}) = 7 \cdot \dfrac{8\left(8^{16}-1\right)}{8-1}$

$\begin{vmatrix} a & 2\left(2^{16}-1\right) & 2^{16}-1 \\ b & 4\left(4^{16}-1\right) & 2\left(4^{16}-1\right) \\ c & 8\left(8^{16}-1\right) & 4\left(8^{16}-1\right) \end{vmatrix}$

$2\begin{vmatrix} a & 2^{16}-1 & 2^{16}-1 \\ b & 4^{16}-1 & 4^{16}-1 \\ c & 8^{16}-1 & 8^{16}-1 \end{vmatrix}$

$2 \times 0 = 0$